Lecture 6: Determinants

- MAT201 Linear Algebra
- What a single number tells you about a matrix

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The determinant of a 2×2 matrix

- For A = [[a, b], [c, d]], det(A) = ad − bc.
- Multiply the diagonal, then subtract the product of the off-diagonal entries.
- Example: det([[3, 2], [1, 4]]) = 3·4 − 2·1 = 10.

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What the determinant measures

- A 2×2 matrix maps the unit square to a parallelogram.
- The area of that parallelogram is |det(A)|.
- A negative determinant means the map also flips orientation.

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Determinant zero

- $\det(A) = 0$ exactly when A squashes the plane onto a line or a point.
- Then the columns of A are linearly dependent.
- A matrix is invertible if and only if $\det(A) \neq 0$.

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A common sign mistake

- det([[a, b], [c, d]]) = ad – bc, not ad + bc.
- The product of the off-diagonal entries is always subtracted.
- Check: det([[4, 1], [2, 3]]) = 12 – 2 = 10.

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Determinants with a variable

- We will often need det(A – λI), where λ is unknown.
- Subtract λ only from the diagonal entries: A – λI = [[a – λ, b], [c, d – λ]].
- Then det(A – λI) = (a – λ)(d – λ) – bc, a polynomial in λ.

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Summary

- det([[a, b], [c, d]]) = ad – bc.
- |det(A)| is the area scaling factor of the map.
- det(A) = 0 means A is not invertible.
- Next lecture: det(A – λI) = 0 gives the eigenvalues.

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